EXPERIMENT 2

NAME: PARAMESWARI P

ROLL NO: 240701378

Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION PAGE

Chunking Analysis of the Instruction Page (Travel Theme)

Chunking is a cognitive strategy that organizes information into small, meaningful units, improving comprehension and memory retention. The travel-themed instruction screen of the Memory Recall Test effectively applies chunking principles as follows:

1. **Clear and Sequential Numbering**

Instructions are presented in a step-by-step manner.

Each step explains a single action (observe, memorize, recall, submit), minimizing cognitive load.

Sequential flow prepares users mentally for the upcoming task.

2. **Logical Grouping of Information**

Observation Phase: Users are informed they will see travel-related icons (luggage, airplane, camera, compass, etc.).

Memorization Phase: Users are guided to focus on remembering the displayed items.

Recall Phase: Users are told to select the items they remember.

Evaluation Phase: Users are informed that their score will be displayed at the end.

3. **Visual Hierarchy and Design Element**

Bold heading: “Memory Recall Test” clearly defines the task.

Bullet-point instructions improve readability.

Travel visuals (passport, suitcase, map icons) provide thematic relevance.

A prominent Start button directs the user intuitively.

4. **Time Expectation Reinforcement**

The instruction page prepares users for a limited memorization time.

This primes the user to stay alert and focused.

5. **Simplicity and Clarity**

Short, direct sentences are used.

No unnecessary technical language.

Instructions are easy to understand even for first-time users.

FRAME 2: CHUNKING / MEMORIZATION PHASE

Analysis of the Memory Recall Test – Chunking Phase Screen

This screen represents the visual encoding phase where users observe and mentally group travel-related items.

1. **Purpose of the Screen**

Allows users to encode visual information into short-term memory.

Encourages chunking by grouping similar travel items.

2. **Key UI Components**

Countdown Timer

Displays remaining memorization time.

Creates urgency and focus.

Progress Bar

Visually represents time passing.

Reinforces time awareness.

Grid of Travel Icons

Includes icons such as suitcase, airplane, camera, compass, sunglasses, map, globe, ship, etc.

Icons are arranged in a grid for easy scanning.

Users naturally apply chunking strategies like:

Grouping by transport (plane, ship, train)

Grouping by accessories (camera, sunglasses, bag)

Grouping by navigation (map, compass, globe)

3. **How Chunking Works Here**

Users scan the grid quickly.

Similar travel items are mentally grouped.

These groups are stored as chunks instead of individual items.

1. **Cognitive & UX Benefits**

Improves short-term memory retention.

Reduces cognitive overload.

Enhances pattern recognition and visual association

FRAME 3: RECALL / SELECTION PHASE

Analysis of the Memory Recall Test – Recall Phase Screen

This screen tests the user’s ability to retrieve stored information from memory.

1. **Purpose of the Screen**

Evaluates memory retention and effectiveness of chunking.

Tests recall accuracy under mild cognitive pressure.

2. **Key UI Components**

Title and Instruction

Reinforces task objective: “Select the items you remember”.

Grid of Icon Options

Contains both previously seen travel icons and new distractor icons.

Encourages careful recall rather than guessing.

Visual Feedback

Selected items are later marked correct or incorrect.

Helps users understand memory strengths and errors.

Submit Button

Clearly placed and easy to access.

Finalizes user selections.

3. How the Recall Phase Works

Users recall memorized travel items.

They select icons they believe were shown earlier.

Distractors test false memory tendencies.

Submission leads to evaluation.

4. Cognitive & UX Benefits

Tests memory accuracy.

Improves discrimination between true and false recall.

Simple interaction reduces confusion.

Gamified visuals keep users engaged.

FRAME 4: RESULT / FEEDBACK PAGE

Analysis of the Memory Recall Test – Result Screen

This screen provides performance feedback and next-step options.

1. **Purpose of the Screen**

Displays recall score clearly.

Motivates users through feedback and rewards.

2. Key UI Components

Score Display

Score shown as a ratio (e.g., 5/7).

Circular score indicator visually highlights performance.

Rating / Stars

Reinforces achievement visually.

Adds a gamified reward element.

Trophy Icon

Symbolizes success and completion.

Encourages positive emotional response.

Action Buttons

Continue: Move to next level.

Restart: Retake the test.

Exit: End the activity.

3. How This Phase Works

System evaluates selections.

Score is calculated and displayed.

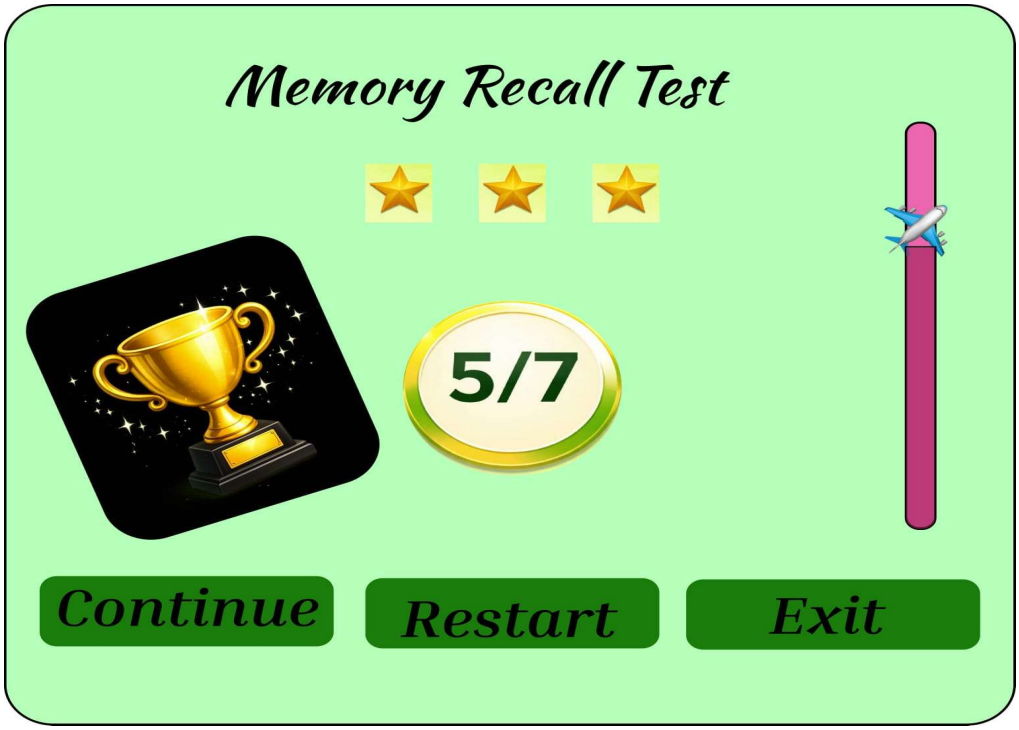
Users choose how to proceed.

4.Cognitive & UX Benefits

Immediate feedback enhances learning.

User control increases engagement.

Reward-based design reduces stress and increases motivation.



Prototype Link:

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